

Topic 2: Nested IF Function

What is Nested IF?

A **Nested IF** means using an IF function inside another IF function.

It is used when there are **multiple conditions** to check.

Example

Instead of only:

- Pass
- Fail

We can assign:

- A Grade
 - B Grade
 - C Grade
 - Fail
-

Syntax

```
=IF(condition1,result1,  
  IF(condition2,result2,  
    IF(condition3,result3,result4)))
```

Demo Table

Create this table:

Student	Marks
Ram	92

Sam 78

Hari 60

Priya 35

fx =IF(B2>=80,"A",IF(B2>=60,"B",IF(B2>=50,"C","Fail")))				
A	B	C	D	E
Student	Marks			
Ram	92	A		
Sam	78	B		
Hari	60	B		
Priya	35	Fail		

Formula

In C2 enter:

```
=IF(B2>=80,"A",IF(B2>=60,"B",IF(B2>=50,"C","Fail")))
```

Result

Student	Marks	Grade
Ram	92	A
Sam	78	B
Hari	60	B
Priya	35	Fail

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How Excel Checks

For Ram (92):

$92 \geq 80 \rightarrow \text{TRUE}$

Result:

A

For Sam (78):

$78 \geq 80 \rightarrow \text{FALSE}$

$78 \geq 60 \rightarrow \text{TRUE}$

Result:

B

For Priya (35):

$35 \geq 80 \rightarrow \text{FALSE}$

$35 \geq 60 \rightarrow \text{FALSE}$

$35 \geq 50 \rightarrow \text{FALSE}$

Result:

Fail

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Real MIS Example

Employee	Sales
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A	25000
---	-------

B	18000
---	-------

C	12000
---	-------

D	7000
---	------

Performance Rules

Sales	Rating
≥ 20000	Excellent
≥ 15000	Good
≥ 10000	Average
< 10000	Poor

Formula:

=IF(B2>=20000,"Excellent",IF(B2>=15000,"Good",IF(B2>=10000,"Average","Poor")))						
A	B	C	D	E	F	G
Employee	Sales					
A	25000	Excellent				
B	18000	Good				
C	12000	Average				
D	7000	Poor				

=====

Formula:

=IF(B2>=20000,"Excellent",IF(B2>=15000,"Good",IF(B2>=10000,"Average","Poor")))

Practice Exercise

Create:

Student	Marks
John	95
David	72

Mary 55

Peter 40

Use:

=IF(B2>=80,"A",IF(B2>=60,"B",IF(B2>=50,"C","Fail")))

Expected Output:

Student	Marks	Grade
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John	95	A
------	----	---

David	72	B
-------	----	---

Mary	55	C
------	----	---

Peter	40	Fail
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fx =IF(B2>=80,"A",IF(B2>=60,"B",IF(B2>=50,"C","Fail")))						
A	B	C	D	E	F	G
Student	Marks					
John	95	A				
David	72	B				
Mary	55	C				
Peter	40	Fail				

=====